

Project 8: Extending Tiny NeRF

Group 4

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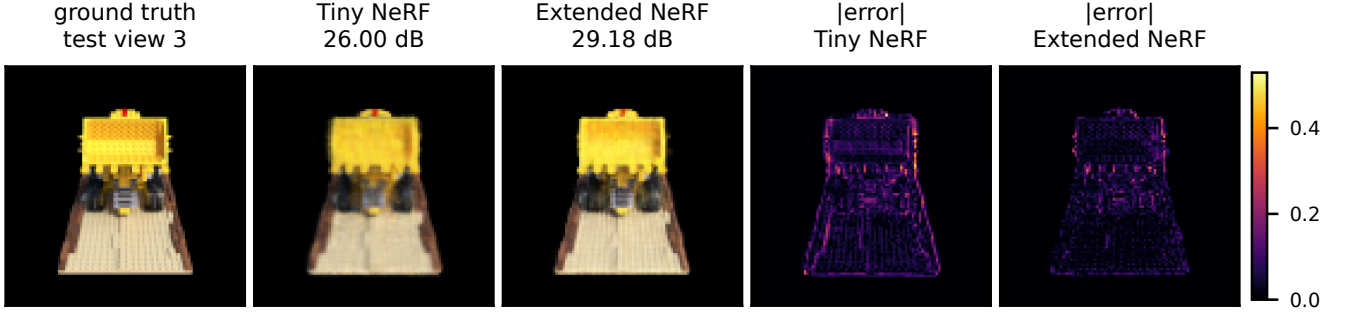


Figure 1: Held-out test view 3, the view ranked in the middle by Extended NeRF’s gain over Tiny NeRF: ground truth, both renderers with their PSNR, and each model’s absolute error on one colour scale.

1. Introduction

Tiny NeRF, from the Week 10 tutorial, learns a scene as a function from a 3D position to a colour and a volume density, and renders a pixel by accumulating that function at 64 evenly spaced depths along the camera ray (Salvado, 2026, p. 71). The full NeRF of Mildenhall et al. (2020) adds two things it lacks: a colour that depends on the viewing direction, and hierarchical sampling, in which a coarse network chooses where a fine network looks. This report adds both, trains on 100 of 106 synthetic 100×100 images of a Lego bulldozer, tests on the other six (images 100 to 105) with PSNR and SSIM (Wang et al., 2004), and isolates each change with an ablation.

2. Implemented changes

Task 1, view-dependent colour. The position, encoded with $L = 6$ frequency bands (39 inputs), passes through two 128-unit layers, and the density is read from them before the direction enters, so the geometry is the same from every camera. A feature layer then meets the unit viewing direction, encoded with $L = 4$ bands (27 inputs; Salvado, 2026, p. 77), and a two-layer head outputs the colour. Ablation A feeds the same network a zero direction: a matte renderer of identical size.

Task 2, hierarchical sampling. A coarse network evaluates $N_c = 32$ stratified depths per ray. Its weights $w_i = T_i(1 - e^{-\sigma_i \delta_i})$, detached from the graph, define a piecewise constant density on the ray, and the starter’s `sample_pdf` inverts its cumulative distribution to draw $N_f = 64$ more depths where the weights are large. A fine network evaluates all 96 sorted depths, and the loss adds the coarse and fine errors so the coarse network keeps learning where the surface is. Ablation B: view-dependent colour, one network, 96 evenly spaced samples.

Protocol. Each model trains for 100 epochs on batches of 1,024 rays with Adam (learning rate decaying from 5×10^{-4} to 5×10^{-5}) and is evaluated with the weights of its final epoch, not of its best test epoch.

3. Results

Extended NeRF reaches a mean test PSNR of 28.44 dB (Table 1), above the 28 dB the task expects, with four of the six views at or above it. It is 3.03 dB above Tiny NeRF and higher on all six views (Figure 1); Figure 2 shows novel views around a full turn. The training curves are nearly flat at the end (Figure 3): in the last ten epochs Extended NeRF gained +0.09 dB, Tiny NeRF +0.01 dB.

Removing view dependence (ablation A) costs 2.03 dB, and Extended NeRF is higher on all six views (by

Model	View	Hier.	Evals	Params	PSNR	SSIM
Tiny NeRF	no	no	64	22,148	25.41	0.894
Ablation A	no	yes	128	96,904	26.41	0.917
Ablation B	yes	no	96	48,452	28.34	0.937
Extended NeRF	yes	yes	128	96,904	28.44	0.940

Table 1: Means over the six held-out views (PSNR in dB). View: direction input. Hier.: coarse and fine networks. Evals: network evaluations per ray.

+1.40 to +2.82 dB). Removing hierarchical sampling (ablation B) costs 0.10 dB, but per view the difference runs from -0.16 to $+0.41$ dB, favouring Extended NeRF only on three of the six views, so the two samplers are not reliably different at the trained budget.

4. Discussion

Rendering quality. Tiny NeRF’s render is visibly softer: the studs on the bulldozer blur together, and its error is brightest along the silhouette and the rim of the base (Figure 1). Extended NeRF resolves the studs, and its remaining error is mostly a thin outline at depth edges. Its turntable frames stay sharper in the same way.

View-dependent appearance. Freezing Extended NeRF’s direction input at one camera’s direction changes its colours by 0.0277 on average (0 to 1 scale) and lowers test view 0 from 26.42 to 21.38 dB (Figure 4, top): the model relies heavily on the direction. To test whether the photographs warrant this, 162 object points away from image edges were projected into the training cameras that see them unoccluded (median 18 per point). Across those cameras, a point’s brightness varies by 0.0199 (median standard deviation), above the 0.0167 that matte ablation A shows from geometry and sampling alone, so the scene does change with viewpoint. Extended NeRF follows that change more closely than ablation A (median correlation 0.69 against 0.40; Figure 4, bottom), but its renders vary by 0.0272, more than the photographs at 75% of points. That excess is what the shape-radiance ambiguity predicts (Zhang et al., 2020): a direction-dependent colour can also absorb geometry errors. NeRF loses more without view dependence (31.01 against 27.66 dB; Mildenhall et al., 2020, Table 2), and its Fig. 4 shows why on this same bulldozer: the tread’s specular reflection is lost.

Hierarchical sampling. The fine samples find the surface: on test view 0’s 3,371 object rays, 84.7% of them fall near it, against 14.3% of 96 uniform samples (near: inside the interval holding the central 90% of the final weights, widened by half its width each side; Figure 5,

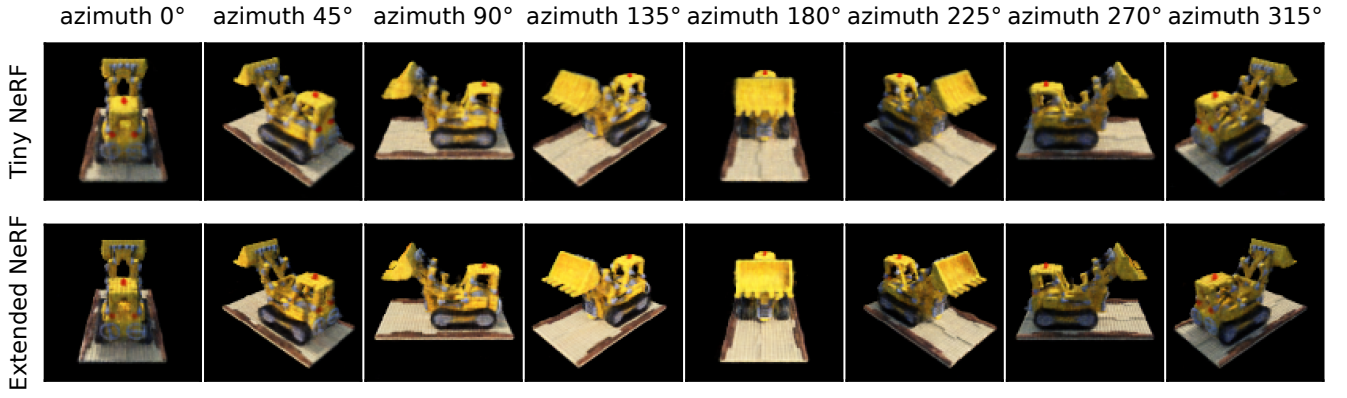


Figure 2: Novel views from a turntable of 40 frames, 30 degrees above the horizontal at radius 4, every 45 degrees of azimuth: Tiny NeRF (top) and Extended NeRF (bottom).

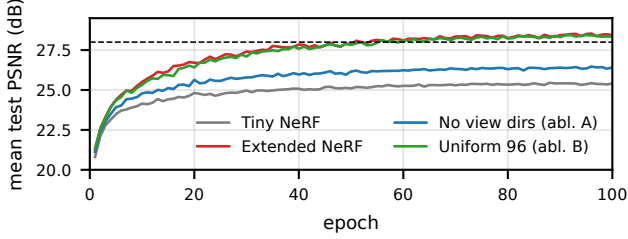


Figure 3: Test PSNR per training epoch, for monitoring only (dashed: 28 dB); all results use the final epoch’s weights.

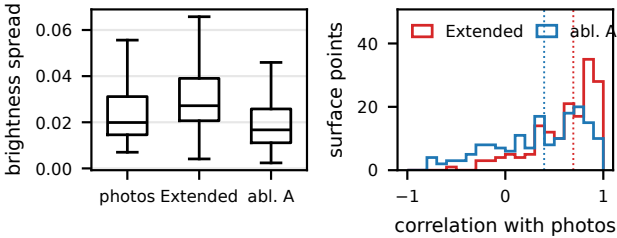


Figure 4: Top: Extended NeRF’s test view 0, with the direction input frozen to view 3’s, and the colour change. Bottom, per object point away from image edges: brightness spread across the cameras that see it (left; quartiles) and correlation with the photographs (right; dotted: medians).

top, shows one ray). That buys little PSNR here: uniform sampling is already near its limit, 28.34 dB at 96 samples and 28.42 dB at 256 (Figure 5, bottom). At equal work (all network evaluations counted), hierarchical sampling trails by 2.61 dB at 32 evaluations per ray, where the coarse pass has only 8 samples to locate thin structures, and by 1.25 dB at 64. It leads only by 0.05 dB at 128 and by 0.16 dB at 256, against 0.95 dB in NeRF’s ablation at an equal 256 (Mildenhall et al., 2020, Table 2).

Overall performance. Extended NeRF’s accuracy has a price: 96,904 parameters against 22,148, 128 network evaluations per ray against 64, 142 against 33 ms per test image on an NVIDIA A16-4Q, and 81.8 against 22.2 minutes of training. Ablation B comes within 0.10 dB of it with 48,452 parameters and 42.9 minutes.

Strengths and limitations. View-dependent colour gives a large, consistent gain for one small head, and importance sampling finds the surface; but hierarchical sampling doubles the networks for no reliable gain here, and

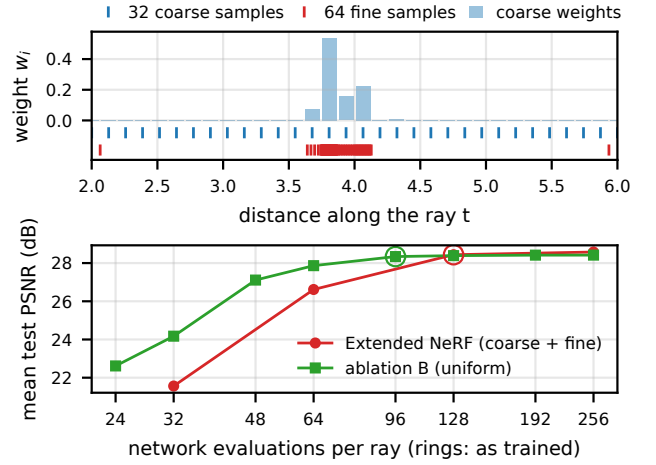


Figure 5: Top: one object ray of test view 0: coarse weights (bars), the 32 coarse samples and the 64 fine samples drawn from those weights (ticks). Bottom: mean test PSNR of the trained models at other sample counts (coarse to fine 1:2).

the direction input can absorb geometry errors. One seed per model cannot separate 0.10 dB from run-to-run variation, six test views are few, and the sweep re-renders rather than retrains.

Recommendations. For a scene this size, view-dependent colour with uniform sampling is the better trade: nearly all the quality at half the parameters and training time. Hierarchical sampling should be retested, with several seeds, where even spacing stops resolving the surface (higher resolution, deeper scenes), and penalising how much the colour varies with direction would show how much of that variation is compensation.

5. Conclusion

Extended NeRF reaches 28.44 dB on the six held-out views, 3.03 dB above Tiny NeRF; view-dependent colour is worth 2.03 dB over its ablation and hierarchical sampling 0.10 dB at the trained budget. The direction input follows real changes in the photographs but exaggerates them, and the coarse-to-fine sampler places samples well but pays only at large budgets here.

References

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